COMPARATIVE ASSESSMENT REPORT

PPR and EXTEDO
A Side-by-Side Comparison

The PPR medicine approval system compared with the EXTEDO eCTD validation and review products used by regulatory authorities

تقرير مقارنة: نظام تسجيل المستحضرات الصيدلانية (PPR) مقابل حلول EXTEDO للتحقق من ملفات eCTD ومراجعتها

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1 Executive summary

This report compares **PPR**, the department's in-house medicine approval system (proof of concept, October 2026), with the **EXTEDO** products that regulatory authorities use to receive, validate and review eCTD dossiers: **EURS** (desktop review system), **EURSnext** (its web companion, sold only to authorities) and **EURSvalidator** (the validation engine).

The two are **different kinds of system**, so they overlap only in part:

- **EXTEDO EURS** is a mature, specialised **validation and dossier-review tool**. By EXTEDO's account, more than 35 authorities use it, including the European Medicines Agency (EMA). EXTEDO also names Oman, Saudi Arabia, Bahrain, Qatar and the UAE among the agencies that use or recommend EURS or EURSvalidator. Its strengths are official, regularly updated validation criteria for each region (GCC included), deep document-review tools (annotation, full-text search, lifecycle navigation) and enterprise scale.
- **PPR** is an **end-to-end registration workflow system**: a company portal, automated eCTD validation, assignment of specialist reviewers per CTD module, information requests, the HOD's decision, and a bilingual approval certificate. Its interface is in Arabic and English, it has an AI dossier assistant, and the department owns it.

> **Key finding.** EXTEDO is stronger where depth and certification matter: validation-rule coverage, document review tools, standards support (NeeS, eCTD 4.0) and scale. PPR is stronger on workflow: it runs the whole registration case from application to certificate, which EURS does not do on its own according to the material reviewed. The PPR validator has 20 rules and checks against a reference DTD. It is a credible start, but it is **not yet equivalent** to the official GCC validation criteria that EURSvalidator applies.

Recommendation (section 9): keep PPR as the registration and workflow system, and close the high-priority gaps in section 8 before going live. Above all, PPR must adopt the official ICH DTD and the GCC Module 1 validation criteria. As a quality check, the department should also benchmark PPR against EURSvalidator on the same set of dossiers, because applicants in the GCC already validate their sequences with EXTEDO tools before submitting. If budget allows, an integration in which PPR manages the case and calls an EXTEDO validator or review tool is a viable alternative.

2 Scope, method and sources

What was compared

- **PPR:** the code in the project repository as of 18 October 2026, including uncommitted work (specialist review teams, automatic assignment, the dossier assistant and the document viewer). The README, data models, workflow, validation rules and tests were reviewed.
- **EXTEDO:** the products offered to regulatory authorities (EURS, EURSnext, EURSvalidator), plus eCTDmanager where the applicant side is relevant. The wider EXTEDOpulse RIM suite (registration, safety, quality and document hubs) is mentioned only where it fills a gap.

Method and limitations

- EXTEDO was assessed from **public material only**: product information sheets, EXTEDO web pages and announcements, and the EMA eSubmission page. No EXTEDO software was installed or tested, and no pricing was available.
- Most EXTEDO claims come from the vendor. Customer counts differ between sources (25, 30 and 35+ authorities). They are quoted as EXTEDO's own claims.
- "Not stated" in this report means that the feature was not described in the material reviewed. It does **not** mean that EXTEDO lacks it. These points should be confirmed with the vendor.
- PPR is a proof of concept. Statements about PPR describe what is in the code, not what has been proven in production.

3 The two systems at a glance

PPR · Pharmaceutical Product Registration

- **Type:** web-based registration workflow system with built-in eCTD validation
- **Built by:** PPR project team for the Pharmacy Department (in-house)
- **Users:** company applicants, specialist reviewers, the lead reviewer, the HOD and an administrator
- **Technology:** Django, Tailwind CSS, SQLite locally and MySQL on PythonAnywhere
- **Languages:** English and Arabic (right-to-left)
- **Validation:** 20 rules in 3 layers (XML schema, business rules, file compliance)
- **Status:** proof of concept with an automated test suite; not yet in production

EXTEDO · EURS, EURSnext, EURSvalidator

- **Type:** commercial eCTD/NeeS validation and review software for authorities
- **Vendor:** EXTEDO (Germany), a Regulatory Information Management (RIM) vendor
- **Users:** agency validators and assessors, including external assessors (EURSnext)
- **Technology:** Windows desktop and server services; EURSnext runs in the browser
- **Languages:** English; Arabic is not mentioned in the material reviewed
- **Validation:** official validation sets per region, including all GCC member states, sold as subscription packages
- **Status:** mature product; selected for EMA use and offered to EU national authorities

Where each system fits in the regulatory process

Process step	PPR	EXTEDO (authority-side products)
Applicant builds the dossier	Not covered. Companies build the eCTD with their own tools.	eCTDmanager (applicant product) builds, validates and publishes dossiers.
Applicant submits	Yes Company portal: create the application, upload the sequence ZIP, submit.	Via gateway Receipt through gateways and portals (CESP connect, EMA Common Repository connector, server monitoring).
Technical validation	Yes Automatic on submission; 20 rules.	Yes EURSvalidator with official regional validation sets.
Scientific review	Yes Assessment per CTD module, recorded in the system.	Yes Viewing, search, annotation and export of annotations for the assessment report.
Case management (assignment, questions, deadlines)	Yes Team assignment, information requests, status tracking, notifications.	Not stated Filtering by regulatory activity; case workflow through the API or other EXTEDO hubs.
Decision and certificate	Yes HOD approves or rejects; certificate with validity period.	Not stated Not part of EURS.

4 Capability coverage matrix

Full available and substantial Partial available but limited None not available Not stated not described in the public material

Capability	PPR	EXTEDO	Comment
VALIDATION			
eCTD v3.2.2 validation	Partial	Full	PPR: 20 rules and a reference DTD. EXTEDO: official criteria sets
GCC Module 1 validation	Partial	Full	PPR checks only that the regional XML exists and is well-formed
NeeS and other formats	None	Full	PPR accepts eCTD only
eCTD 4.0 readiness	None	Partial	EXTEDO says it is "prepared for" eCTD 4.0
PDF compliance checks	Full	Full	PPR: version, security, JavaScript, fonts, size
Validation report export	Partial	Full	PPR: on screen and CSV. EXTEDO: short and long PDF reports, auto-save, email
REVIEW			
CTD tree and dossier viewer	Full	Full	
Lifecycle (cumulative) view	Partial	Full	PPR validates lifecycle operations but browses sequence by sequence
PDF annotation	None	Full	EXTEDO annotates without changing the originals
Full-text search	Partial	Full	PPR: search through the assistant, within one dossier
AI dossier assistant	Full	Not stated	PPR: cited answers; the live LLM path is not yet tested
WORKFLOW AND ADMINISTRATION			
Applicant portal	Full	None	EXTEDO receives dossiers through gateways
Specialist assignment per module	Full	Not stated	PPR: by the HOD, or automatic by workload
Information requests	Full	Not stated	
Decision and certificate	Full	None	
KPIs and dashboards	Partial	Not stated	PPR: HOD dashboard and average days to decision
Audit trail	Full	Full	
Arabic interface	Full	Not stated	
PLATFORM			
Integration API	None	Full	EXTEDO: REST API, Active Directory, document management systems
Scale and parallel processing	Partial	Full	EXTEDO: 400+ users, distributed services
Formal system validation	None	Full	EXTEDO: "installed and validated within days"

5 Detailed comparison

Area	PPR	EXTEDO
Primary purpose	Manage a marketing authorisation application from submission to certificate, with validation built in.	Validate, accept, import, review and maintain eCTD and NeeS submissions inside an authority.
Submission intake	Companies upload one ZIP per sequence (0000, 0001, …) through the portal. Safety checks on the ZIP: no zip-slip, symlinks, encryption or zip bombs.	Import from gateways: CESP connect (automatic import of CESP/FTP receipts), CR Connect (EMA Common Repository), and server monitoring that unzips and imports automatically.
Application types	New drug, generic, variation, renewal. The modules required for each type are enforced by rule BR-06.	Any standard submission format and regulatory activity; filtering by activity and by product status.
Validation engine	20 rules in 3 layers, each rule a small class, so new rules are easy to add. Runs synchronously. Any error fails the sequence; warnings let it pass.	EURSvalidator applies official validation sets that EXTEDO keeps current in cooperation with authorities. The validation set and DTD are selected automatically. Batch validation of all pending submissions.
Validation results	Stored per run with findings; shown as a report with CSV export.	Short PDF report (errors) and long PDF report (errors, warnings and details); reports are saved automatically and can be emailed.
Dossier navigation	CTD tree for each accepted sequence, with links to the documents (served through access-controlled views).	Navigation through the whole lifecycle of a product, envelope information, and table-of-contents elements.
Document review tools	PDFs open in the browser. The assistant can search and read documents and cites `sequence/path p.N`.	Annotate, highlight and hyperlink PDFs without changing the originals; export annotations to support the assessment report; full-text search across documents, envelopes, TOC elements and annotations.
Review organisation	The HOD sets each reviewer's specialties (M1–M5) and assigns one specialist per module plus a lead reviewer, or the team is assembled automatically by workload. Reviewers write assessments only for their own modules. The lead sends questions and the recommendation once every module is assessed.	Parallel work by many users. Not stated: per-module assignment, recommendation or decision workflow.
Communication with applicants	Information requests with due dates; the company answers with a validated response sequence. In-app and email notifications.	Not stated for EURS (handled outside the review tool).
Decision	The HOD approves (certificate with a validity period), rejects, or returns the application to the lead.	Not part of EURS.

Area	PPR	EXTEDO
AI assistance	Dossier assistant on every page: it knows the current application and document, uses read-only tools limited to what the reader may see, and runs in offline mode (search passages) without an API key.	Not stated in the material reviewed.
Security and access	Role-based access, documents served only to authorised users, audit trail in `StatusHistory`.	Extensive audit trail of user activity; Active Directory integration; optional storage in a document management system (Documentum, FileNet).
Integration	None yet (Django admin for administration).	REST API, command-line automation engine, Web Broker for task queues, connectors for portals and document management systems.
Scale	Single web application on PythonAnywhere; validation and the assistant run inside the web request (no background workers on basic plans).	Scales beyond 400 users; background Windows services and load balancing for long imports and validations.
Deployment	Web; PythonAnywhere with MySQL. No client software.	EURS: desktop client with server services, installed on site. EURSnext: browser access, suited to external assessors.
Language	English and Arabic, with a full right-to-left layout.	English; Arabic not mentioned.
Cost and ownership	Owned by the department; hosting costs and staff time. Can be changed freely.	Commercial licences and validation-set subscriptions, priced per region (prices not public). Updates to standards are the vendor's job.
Maturity and references	Proof of concept from October 2026 with automated tests; no production use yet.	Selected for EMA use and offered to EU national authorities; EXTEDO reports more than 35 authority customers, including GCC agencies.

6 eCTD technical validation in depth

Technical validation was the HOD's core requirement, so it is compared in more detail. Both systems follow the same principle as the FDA and EMA: an automated technical check must pass before scientific review starts.

PPR validation rules

Layer	Rules	What is checked
1. XML schema	SCH-01 to SCH-04	`index.xml` exists, is well-formed and is valid against the eCTD 3.2 DTD; the regional Module 1 XML exists and is well-formed.
2. Business rules	BR-01 to BR-09	Sequence numbering; `index-md5.txt`; every leaf points to an existing file with a matching MD5; no unreferenced files; required modules for the application type; lifecycle operations (new, replace, append, delete) resolve to earlier accepted sequences; ICH file naming and path length; DTD present.
3. File compliance	FC-00 to FC-07	Safe ZIP; accepted file formats (DOCX only in Module 1); PDFs not password-protected; PDF version 1.4–1.7; no JavaScript or launch actions; embedded fonts (warning); size limit; no empty files.

How this compares with EURSvalidator

Where PPR is on par

- Checksums of the backbone and of every leaf
- Broken references and unreferenced files
- Lifecycle operations against earlier sequences
- PDF security, version, JavaScript and fonts
- File naming and path-length rules
- Clear separation of errors and warnings

Where EURSvalidator goes further

- The official GCC Module 1 criteria: the regional XML is checked in content, not only for well-formedness
- The official ICH DTD, rather than the reference copy that PPR currently bundles
- Validation sets kept current with every specification change, in cooperation with authorities
- Automatic selection of the validation set and DTD
- NeeS, and preparation for eCTD 4.0
- PDF reports suitable for sending to applicants

> **Practical risk.** GCC applicants typically validate their sequences with EXTEDO tools (EURSvalidator or eCTDmanager) before submitting. If PPR applies different or fewer criteria, a dossier that passed the applicant's check could fail in PPR, or a dossier with a real defect could pass. Aligning PPR's rules with the official GCC validation criteria, and testing PPR against the same dossiers, is therefore the most important step before go-live.

7 Strengths and weaknesses

PPR · strengths

- Covers the complete case, from application to certificate, in one system
- Review by specialists per CTD module, with a lead reviewer and automatic assignment by workload
- Arabic and English, built for the department's own process
- AI assistant that answers from the dossier and cites page references
- Owned by the department: no licence fees, and it can be changed quickly
- Web-only, so there is no client software to install

PPR · weaknesses

- The validation criteria are not yet the official GCC set, and the DTD is a reference version
- No PDF annotation and no cumulative lifecycle view
- Synchronous processing limits dossier size and the number of users
- No formal computer system validation, and no proof yet in production
- Keeping up with specification changes falls on the team
- Relies on a small team for support

EXTEDO · strengths

- Validation sets aligned with authorities, covering the GCC and kept current by the vendor
- Proven at EMA and many national authorities
- Rich review tools: annotation, full-text search, lifecycle navigation
- Enterprise features: API, Active Directory, document management integration, 400+ users
- Vendor support, training and consulting (Agency Services)

EXTEDO · weaknesses

- No applicant portal, case workflow, decision or certificate in EURS itself
- Licence and subscription costs (not public), plus local partner fees
- Mainly a Windows desktop client; full web access only through EURSnext
- Arabic interface not mentioned
- Customisation depends on the vendor

8 Gap analysis: what PPR needs for production

The items below would bring PPR closer to the level of the EXTEDO tools for the parts the department needs. Effort is a rough estimate for the current team.

#	Gap	Action	Priority	Effort
1	Reference DTD instead of the official one	Install the official ICH eCTD v3.2.2 DTD (the `ECTD_DTD_PATH` setting already exists) and re-run validation on all sample dossiers.	**High**	Small
2	GCC Module 1 checked only for well-formedness	Implement the GCC eCTD specification (version 1.5, per the sources consulted) and its validation criteria as new rules: regional DTD, envelope fields, Module 1 structure.	**High**	Medium
3	No benchmark against a reference validator	Validate the same set of real and broken dossiers with PPR and EURSvalidator; record and resolve every difference.	**High**	Small
4	Synchronous validation on PythonAnywhere	Move validation and text indexing to a scheduled task or queue, and confirm the upload size limits for large dossiers (often several GB).	**High**	Medium
5	No formal system validation	Write validation documentation (requirements, IQ/OQ/PQ, traceability) and complete a security review before go-live.	**High**	Medium
6	No cumulative lifecycle view	Add a "current view" of the dossier that applies replace, append and delete operations across all accepted sequences.	**Medium**	Medium
7	Validation report only on screen and as CSV	Add a printable PDF validation report (short and long versions) to send to applicants.	**Medium**	Small
8	No review annotations	Add notes linked to document pages, kept separate from the original files, and export them into the assessment report.	**Medium**	Large
9	Assistant tested only in offline mode	Test the live LLM path with a real key, and agree on a data-protection position for sending dossier text to an external model.	**Medium**	Small
10	Search limited to one dossier	Add full-text search across dossiers for reviewers and the HOD, using the existing `DocumentText` cache.	**Low**	Medium
11	No integration interface or single sign-on	Add Active Directory or Ministry single sign-on, and a small API if other systems need data.	**Low**	Medium
12	No eCTD 4.0 or NeeS support	Watch GCC plans for eCTD 4.0; design the validation engine so that a second backbone format can be added.	**Low**	Large

9 Options and recommendation

Option	Description	Advantages	Disadvantages
A. PPR on its own	Continue PPR and close the gaps in section 8.	Lowest cost; full control; Arabic; complete workflow.	The team must build and maintain the GCC criteria; higher validation risk at first.
B. EXTEDO on its own	License EURS/EURSnext and EURSvalidator; manage the workflow outside the system (email, paper, spreadsheets).	Proven validation and review tools; vendor support.	No portal, assignment, decision or certificate; licence costs; English only.
C. Hybrid	PPR manages the case, portal and decisions; an EXTEDO validator or EURS is called through its API, or used alongside PPR for validation and review.	Best validation and review depth together with the department's own workflow.	Licence costs plus integration work; two systems to support.

Recommendation. Proceed with **option A** and keep **option C** open:

- Complete gaps 1 to 5 before go-live, starting with the official ICH DTD and the GCC Module 1 criteria.
- Benchmark PPR against EURSvalidator (gap 3). A trial or evaluation licence could be requested from EXTEDO or its regional partner for this purpose.
- If the benchmark shows differences that are costly to close, or if dossier volumes grow beyond what PythonAnywhere can handle, move to option C and let PPR call an external validator while keeping its workflow, Arabic interface and assistant.
- Ask EXTEDO to confirm the points marked "Not stated" in this report, and to provide a price for a GCC validation-set subscription, so the options can be compared on cost.

A Sources

Retrieved 18 October 2026. Vendor material is quoted as the vendor's own claims.

- PPR project repository: `README.md`, `apps/applications`, `apps/validation`, `apps/assistant`, `tests/` (state as of 18 October 2026).
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